

Retail Sales Performance Analytics

Business Report • Tata Data Visualization Job Simulation

1. Business Problem & Objectives

Business Objective

Develop a centralized sales analytics solution to evaluate revenue performance, customer behaviour, and operational performance, supporting data-driven commercial and operational decision-making.

Key Objectives

- Assess revenue and sales performance across markets, products, and channels.
- Analyse customer purchasing behaviour and engagement.
- Identify operational patterns and performance drivers.
- Provide actionable insights to support CEO and CMO decision-making.
- Establish a structured **RAW** → **CLEAN** → **ANALYTICS** → **BI** workflow for reliable reporting.

Analysis Period

December 2010 – December 2011

Reporting Scope Note

December 2011 represents a partial reporting period. Therefore, December should be interpreted cautiously when evaluating monthly trends, seasonality, and period-over-period performance.

2. Raw Dataset Overview

Metric		Value
Raw Records		541,909
Variables		8
Products		4,223
Countries		38
Dataset Type		Online Retail Transactions

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010 08:26	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850.0	United Kingdom

3. Data Quality Assessment

A comprehensive assessment was performed on the raw dataset before any transformations were applied. The assessment included:

- Dataset structure validation
- Schema verification
- Missing value analysis
- Duplicate record detection
- Data type validation
- Statistical profiling
- Inconsistency Business rule investigation

3.1 Data Quality Findings

The initial assessment identified several data quality issues that should be addressed before proceeding with data cleaning and business analysis. Although the dataset has a complete schema, several data-quality issues were identified that may affect the accuracy of revenue, customer, product, and operational reporting if left unresolved.

Issue	Key Finding	Action
Invoice Date Stored as Text	<code>InvoiceDate</code> is stored as a text field, preventing time-based analysis.	Convert to <code>datetime</code> .
CustomerID Data Type	Customer identifiers contain no invalid decimal values. The <code>float64</code> type is caused by missing values rather than incorrect data.	Convert to nullable <code>Int64</code> during cleaning.
Missing Product Descriptions	1,454 records (0.27%) have missing descriptions. Investigation confirmed all had zero unit price and did not contribute to revenue.	Remove from the analytical dataset.
Missing CustomerID	135,080 transactions (24.93%) have missing CustomerIDs. 98.9% originate from the United Kingdom and contribute 14.85% of gross total revenue.	Retain for revenue analysis; exclude from customer-level analysis.
Duplicate Records	5,268 duplicate records (0.97%) were identified across 1,933 invoices. Exact duplicate rows were identified and removed based on identical transaction attributes.	Remove duplicate records.

Negative Quantity	474 non-cancellation records were identified as internal inventory and administrative adjustments with zero unit price and no revenue impact.	Remove operational adjustment records; retain legitimate customer returns.
Negative Unit Price	Only 2 records contained negative unit prices. Manual inspection confirmed they were bad debt accounting adjustments, not retail sales.	Remove from the analytical dataset.
Zero Unit Price	582 zero-price transactions were identified. Only operational records (e.g., check, found, adjustment, Manual, Amazon) were excluded, while genuine product transactions were retained.	Remove operational records; retain legitimate zero-price customer transactions.

4. Methodology

4.1 Project Workflow

The project follows a structured analytics workflow to ensure data consistency before visualization.



4.2 Data Preparation

All data-cleaning and transformation rules were validated against the identified data-quality issues before implementation to ensure analytical accuracy and business consistency.

The data preparation process included:

- Removing **5,268 duplicate transaction records** identified across **1,933 invoices**.
- Converting **InvoiceDate** from text to **datetime** format for time-based analysis.
- Converting **CustomerID** from **float64** to nullable **Int64** format to correctly handle missing customer identifiers.
- Removing **1,454 records with missing product descriptions** where the transactions had zero unit price and no revenue impact.
- Retaining transactions with **missing CustomerIDs** for revenue and transaction-level analysis while excluding them from customer-level analysis.
- Identifying and removing **474 non-cancellation negative-quantity records** associated with internal inventory and administrative adjustments, while retaining legitimate customer returns.
- Removing **2 records with negative unit prices** confirmed as bad debt accounting adjustments rather than genuine retail transactions.
- Reviewing **582 zero-unit-price transactions** and removing operational records while retaining legitimate customer transactions where applicable.
- Preserving **cancellation and return transactions** where they represented genuine customer or operational activity required for revenue-loss and operational analysis.
- Creating derived fields and business metrics required for **revenue, customer, product, geographic, and operational analysis**.
- Validating the cleaned dataset after transformation to ensure that the resulting records remained suitable for downstream analysis and reporting.

Each transformation was first validated in Jupyter Notebook before being automated within the ETL pipeline.

4.3 Analytical Layer Development

Following data preparation, analytical datasets were developed to optimize reporting performance.

Current analytical transformations include:

- Creating transaction-level **Revenue** using **Quantity × UnitPrice**.
- Creating time-intelligence fields including **Invoice Year, Invoice Month, Month Name, Quarter, Month Start, and Year-Month** to support monthly, quarterly, and seasonal analysis.
- Creating a **Customer Type** classification to distinguish between **Registered Customers** and **Guest Customers** based on CustomerID availability.
- Creating an **Order Type** classification to distinguish between **Sales** and **Cancellations** using the invoice-number structure.
- Creating **Invoice Revenue** by aggregating transaction revenue at the invoice level.
- Creating **Basket Size** using the number of unique products purchased within each invoice.
- Validating invoice-level metrics using individual invoice records and descriptive statistics across **23,970 unique invoices**.
- Creating **Purchase Frequency** by calculating the number of unique invoices associated with each customer.
- Classifying customers as **Repeat Customers** or **One-Time Customers** based on purchase frequency.

- Retaining cancellation and return activity within the reporting dataset to support revenue-loss and operational analysis.

4.4 Key Summary

KPI	Value
Gross Revenue	£10.64M
Cancelled Revenue	£-894.0K
Total Revenue	£9.75M
Registered Customer Revenue	£8.28M
Guest Revenue	£1.47M
Total Orders	24K
Total Customers	4K
Average Order Value	408.61
Cancellation Order Rate	16.08 %
Repeat Customer Rate	65%
Average Purchase(complete order) Frequency	4.27

4.4 Power BI Semantic Model

A simplified **star-schema model** was implemented in Power BI to support consistent filtering, time-based analysis, and business reporting. The model uses **SalesAnalytics** as the central transaction table, supported by dedicated customer and date dimensions.

Table Name	Primary Key	Foreign Key	Data Grain	Business Purpose
SalesAnalytics	—	—	Invoice × Product	Stores transaction-level sales records used for revenue, product, customer, and geographic analysis.
DimCustomer	CustomerID	CustomerID	Customer	Stores unique customer attributes to support customer segmentation and purchasing behavior analysis.

DimDate	Date	InvoiceDate	Date	Supports time intelligence, trend analysis, and calendar-based reporting.
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The model relationships were established as **one-to-many (1:*)** relationships from the dimension tables to **SalesAnalytics**:

- **DimCustomer[CustomerID] → SalesAnalytics[CustomerID]**
- **DimDate[Date] → SalesAnalytics[InvoiceDate]**

The final model was intentionally kept simple to provide the required analytical capability without introducing unnecessary additional fact tables or model complexity.

6. Detailed Analysis

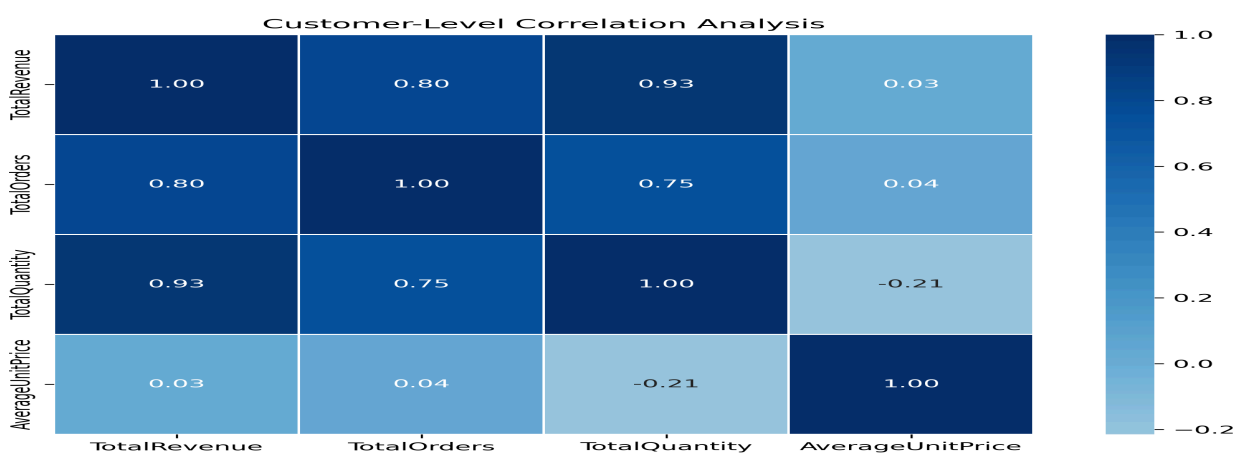
6.1 Exploratory Data Analysis

Summary statistics

	count	mean	std	min	25%	50%	75%	max
Quantity	534593.0	9.959769	217.103446	-80995.0	1.00	3.0	10.00	80995.0
UnitPrice	534593.0	4.691788	95.038019	0.0	1.25	2.1	4.13	38970.0
Revenue	534593.0	18.234678	380.780408	-168469.6	3.75	9.9	17.46	168469.6

- Revenue is highly right-skewed. The median transaction-line revenue is 9.90, compared with a mean of 18.23, while the maximum reaches 168,469.60, showing that a small number of high-value transactions pull the average upward.
- Most transaction lines generate relatively low revenue. 75% generate 17.46 or less, confirming that the majority of sales activity is concentrated in lower-value transactions.
- Negative Quantity and Revenue values reflect reversal activity. These transactions represent returns or cancellations that remain in the analytical dataset and should be evaluated separately when measuring sales performance.

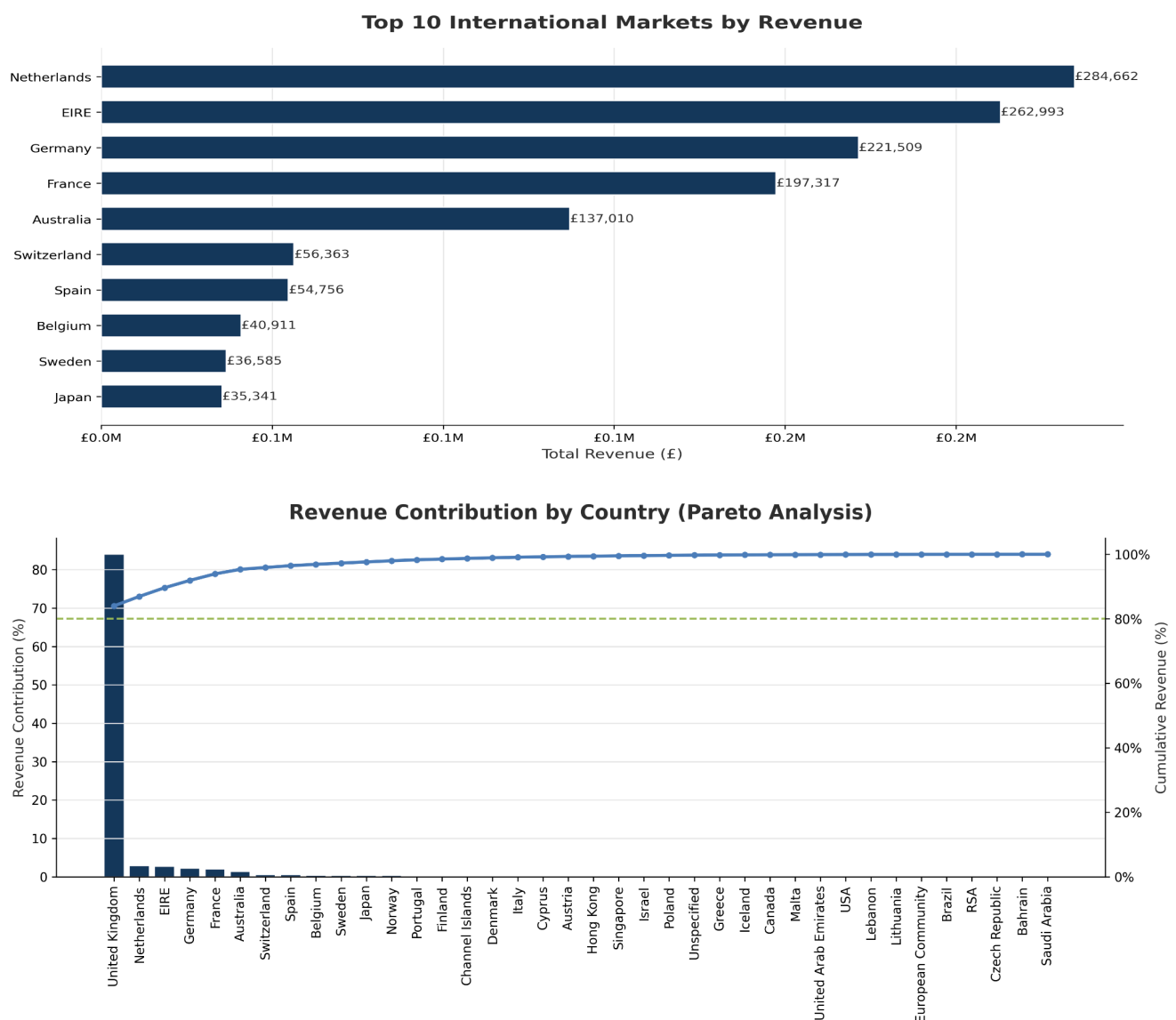
Correlation



- Revenue is strongly associated with purchasing volume. Total Revenue has a strong positive correlation with Total Quantity (0.93), indicating that customers purchasing more units tend to generate higher revenue.
- Repeat purchasing is also strongly associated with revenue. Total Revenue shows a strong positive correlation with Total Orders (0.80), suggesting that customers who place orders more frequently tend to contribute more revenue.
- Average unit price shows a minimal relationship with customer revenue. The correlation between Average Unit Price and Total Revenue is only 0.03, indicating that higher customer revenue is not strongly associated with purchasing higher-priced products.

Research Questions & Key Findings

1. Revenue Concentration and Market Expansion



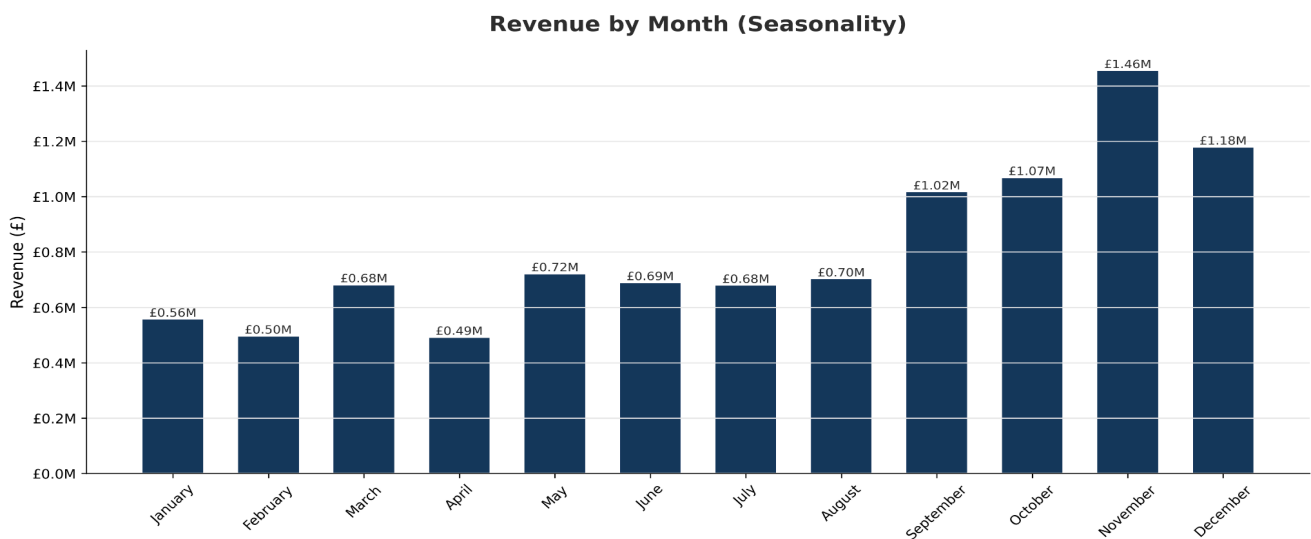
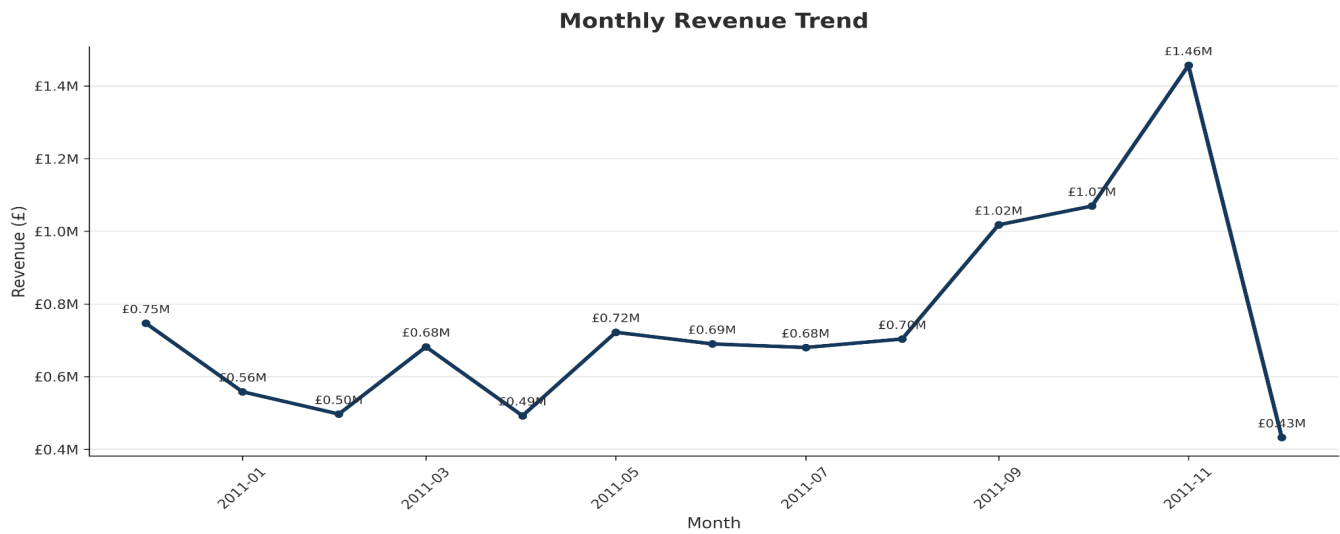
Market Concentration

- The United Kingdom generated £8.19M in revenue, contributing approximately 84.0% of total revenue, indicating a high concentration of revenue within a single market.

International Growth Opportunities

- Among international markets, the Netherlands (£284.7K), EIRE (£263.0K), Germany (£221.5K), and France (£197.3K) generated the highest revenue, making them the strongest current overseas markets.

2. Seasonal Revenue Performance and Peak Trading Periods



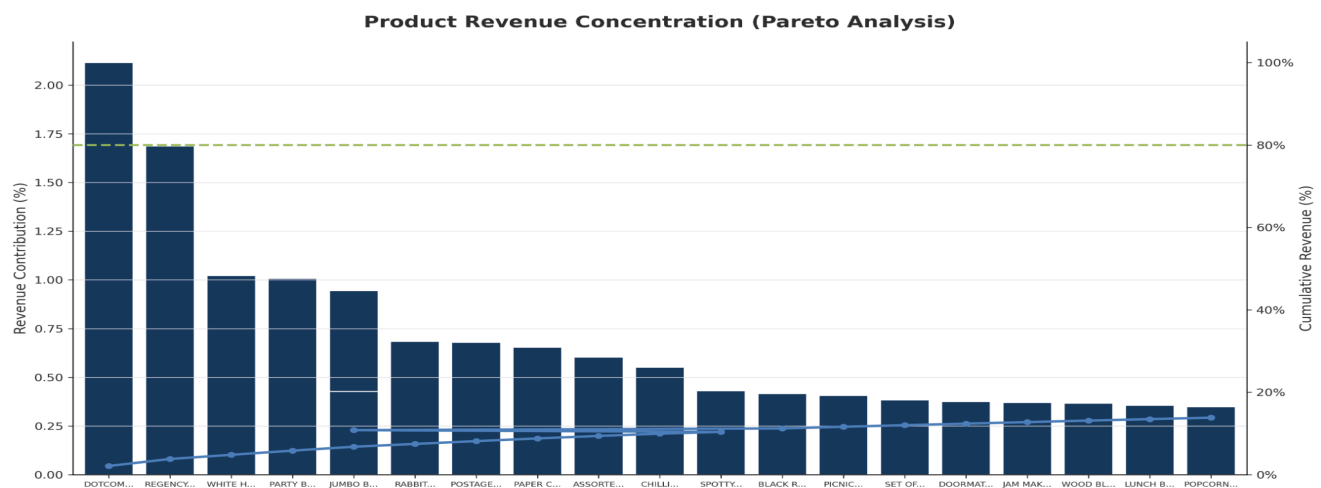
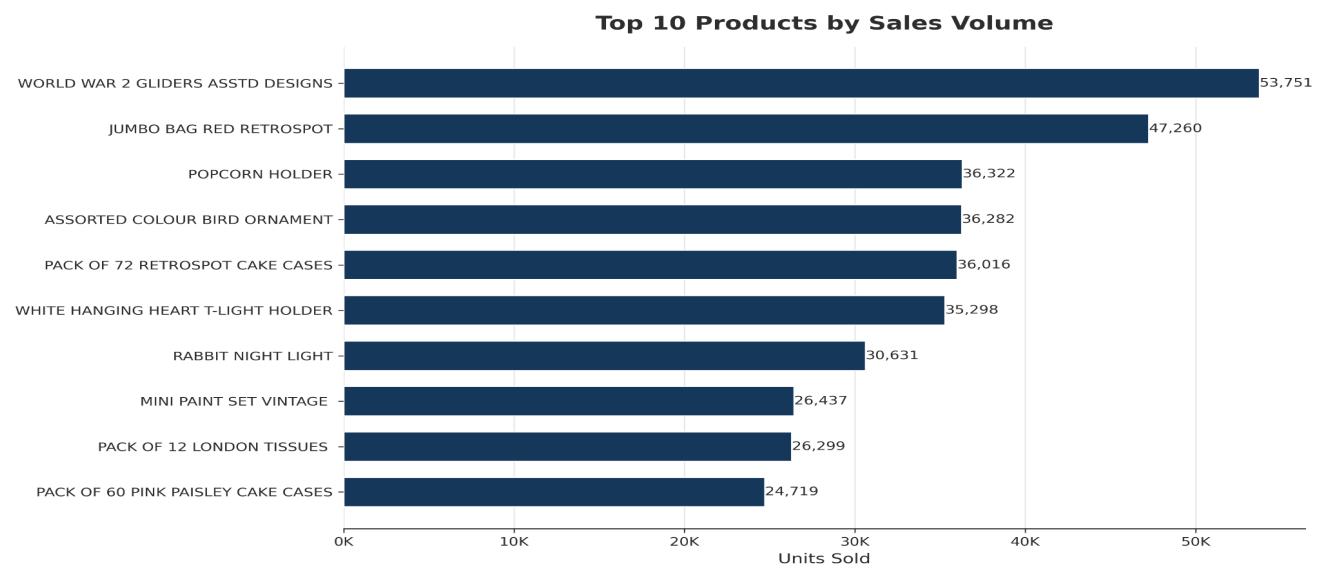
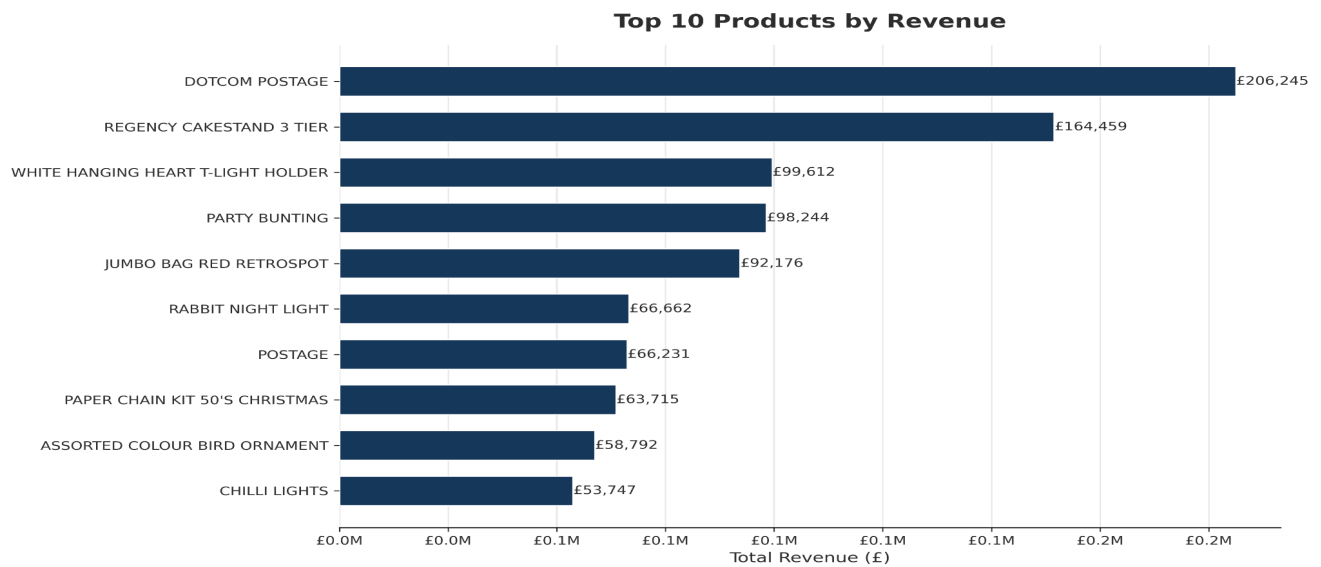
Revenue Trend

- Monthly revenue remained relatively stable between £0.49M and £0.72M from January to August before accelerating sharply to over £1.0M during September–December.

Seasonal Performance

- November recorded the highest full-month revenue at £1.46M. December revenue reached £1.18M despite representing a partial reporting period, so it should not be compared directly with full months.

3. Product Revenue Contribution and Sales Concentration



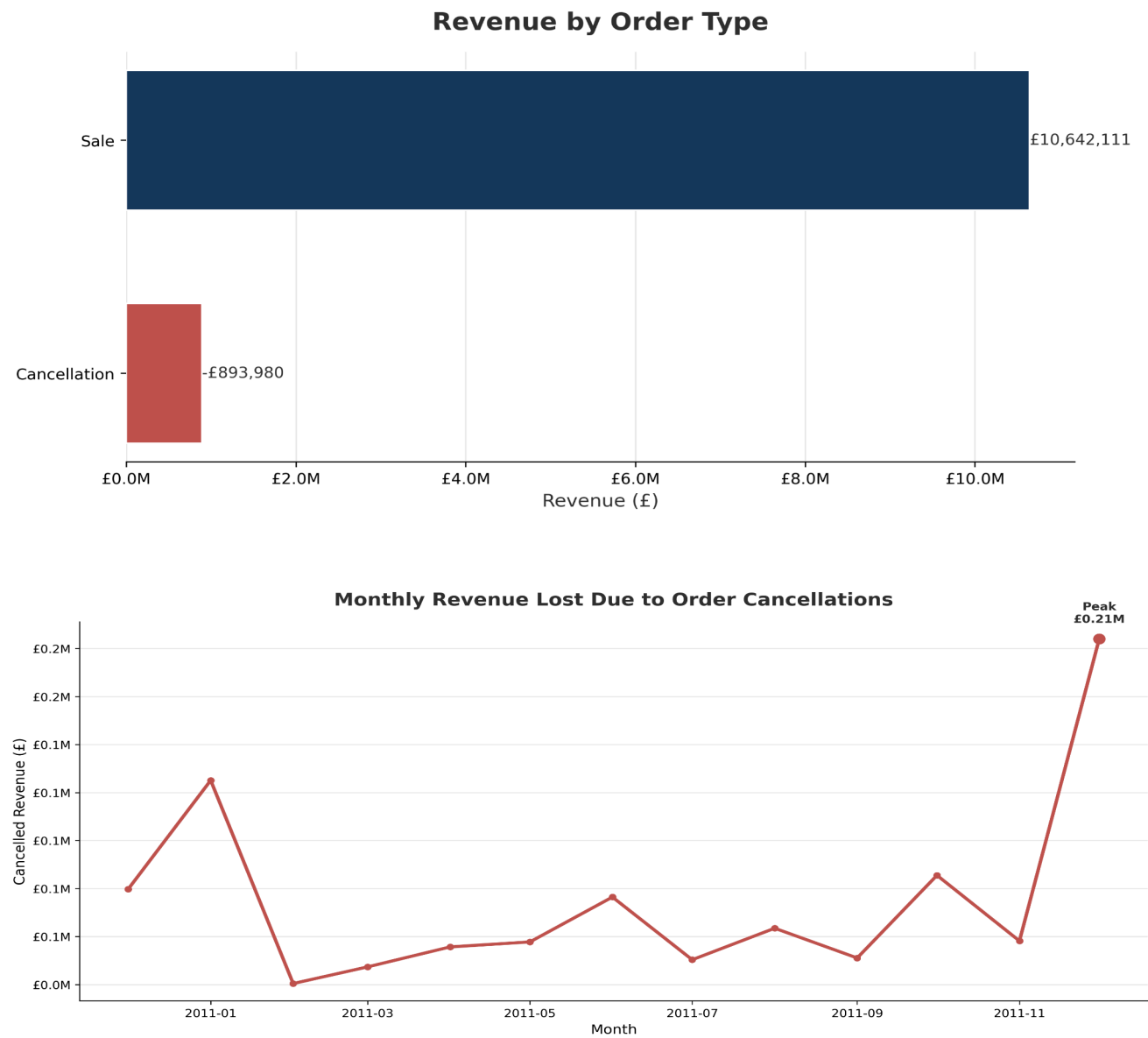
Product Performance

- WORLD WAR 2 GLIDERS ASSTD DESIGNS recorded the highest sales volume at 53,751 units, while DOTCOM POSTAGE generated the highest revenue at £206.2K, showing that the products driving unit sales are not necessarily the same as those generating the most revenue.

Revenue Concentration

- The top 10 products contribute approximately 9.9% of total revenue, indicating that revenue is distributed across a broad product portfolio rather than being heavily dependent on a small number of individual products.

4. Revenue Leakage from Order Cancellations



Cancellation Impact

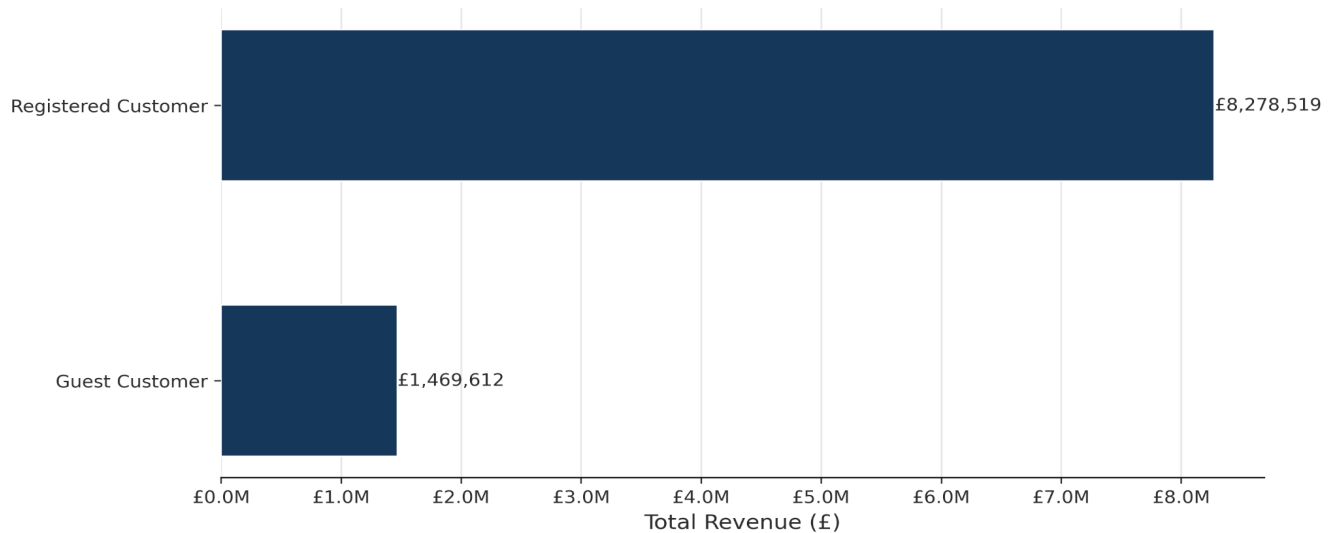
- Order cancellations resulted in approximately £894.0K in lost revenue, equivalent to 8.4% of gross sales revenue, making cancellations a significant source of revenue leakage.

Cancellation Trend

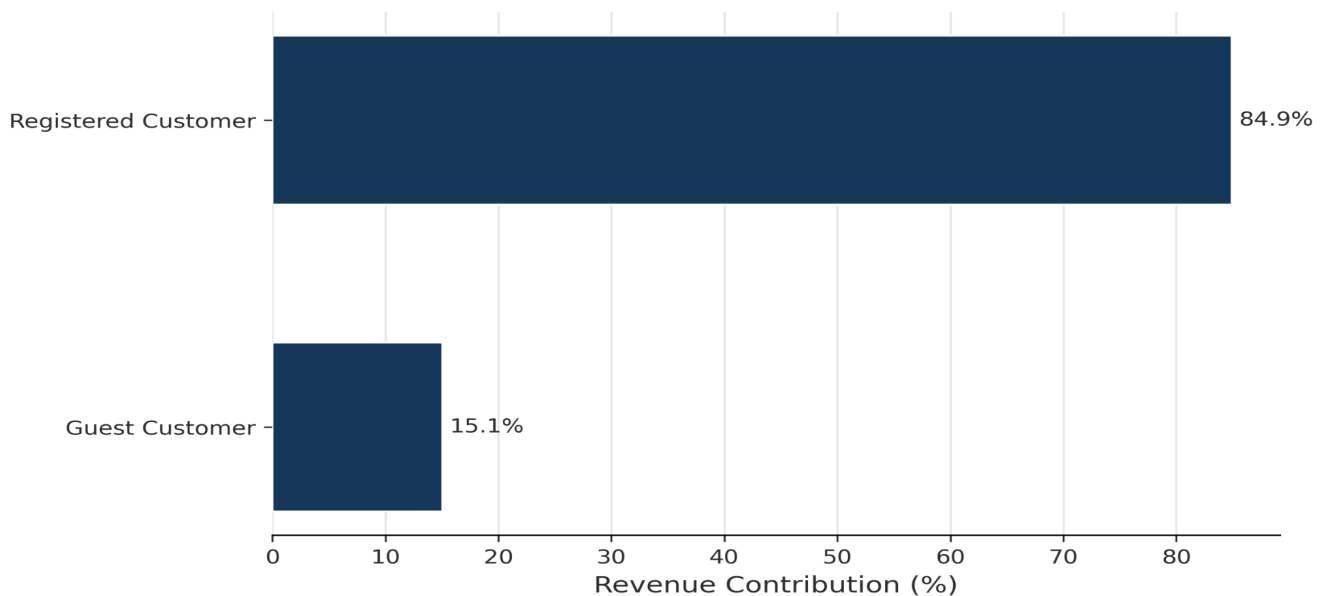
- Cancellation-related revenue loss increased sharply during the year-end period, peaking at approximately £0.21M in December, indicating greater cancellation risk during the highest-demand sales period.

5. Customer Value and Purchase Engagement

Revenue by Customer Segment



Revenue Contribution by Customer Segment



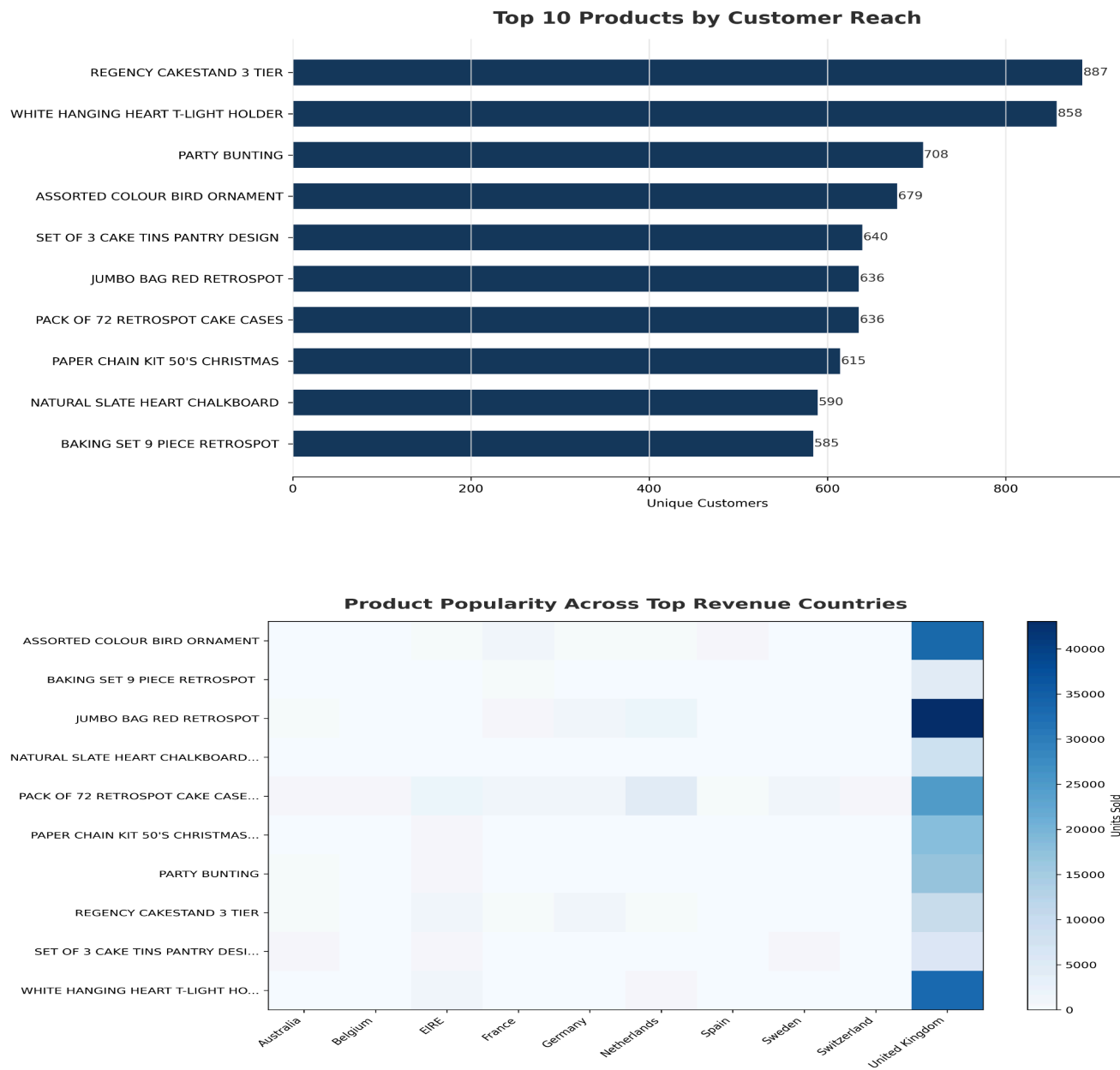
Customer Revenue Distribution

- Registered Customers generated £8.28M (84.9%) of total revenue, compared with £1.47M (15.1%) from Guest Customers, making the business highly dependent on its registered customer base.

Customer Value

- Registered Customers contributed 5.6× more revenue than Guest Customers and placed an average of 4.3 purchases per customer, demonstrating the importance of repeat customers to overall revenue performance.

6. Product Reach and Market-Specific Preferences



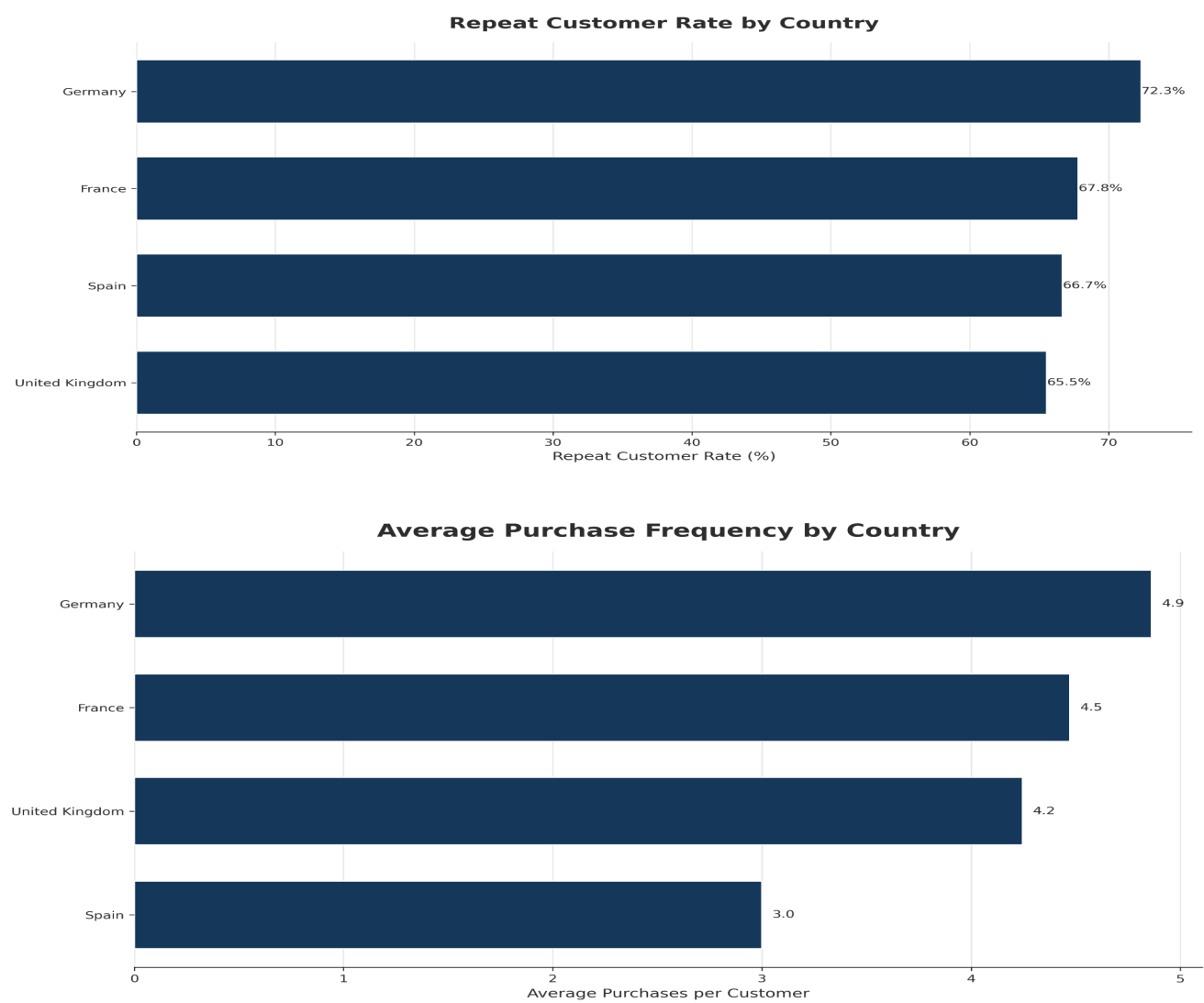
Product Popularity

- REGENCY CAKESTAND 3 TIER reached the largest registered customer base, purchased by 887 unique customers, followed by WHITE HANGING HEART T-LIGHT HOLDER (858) and PARTY BUNTING (708), indicating broad customer appeal.

Country Purchasing Patterns

- The United Kingdom records the highest sales volume across nearly all high-reach products, with products such as JUMBO BAG RED RETROSPOT and ASSORTED COLOUR BIRD ORNAMENT selling substantially more units than in international markets.

7. Customer Engagement and Repeat Purchasing Across Markets



Customer Engagement

- Germany recorded the highest average purchase frequency at 4.9 completed purchases per customer, followed by France at 4.5 and the United Kingdom at 4.2, among countries with at least 30 identifiable customers.

Customer Loyalty

- Germany also recorded the highest repeat customer rate at 72.3%, followed by France (67.8%), Spain (66.7%), and the United Kingdom (65.5%).

8. Customer Purchasing Behaviour and Marketing Opportunities

Customer Purchasing Behavior

	CustomerType	AverageBasketSize	AverageOrderValue	AveragePurchaseFrequency
0	Guest Customer	79.733213	881.062140	<NA>
1	Registered Customer	18.098968	373.091145	4.271722

Customer Purchasing Behavior

- Guest Customers place substantially larger orders, with an average basket size of 79.7 items and an average order value of £881, compared with 18.1 items and £373 for Registered Customers.

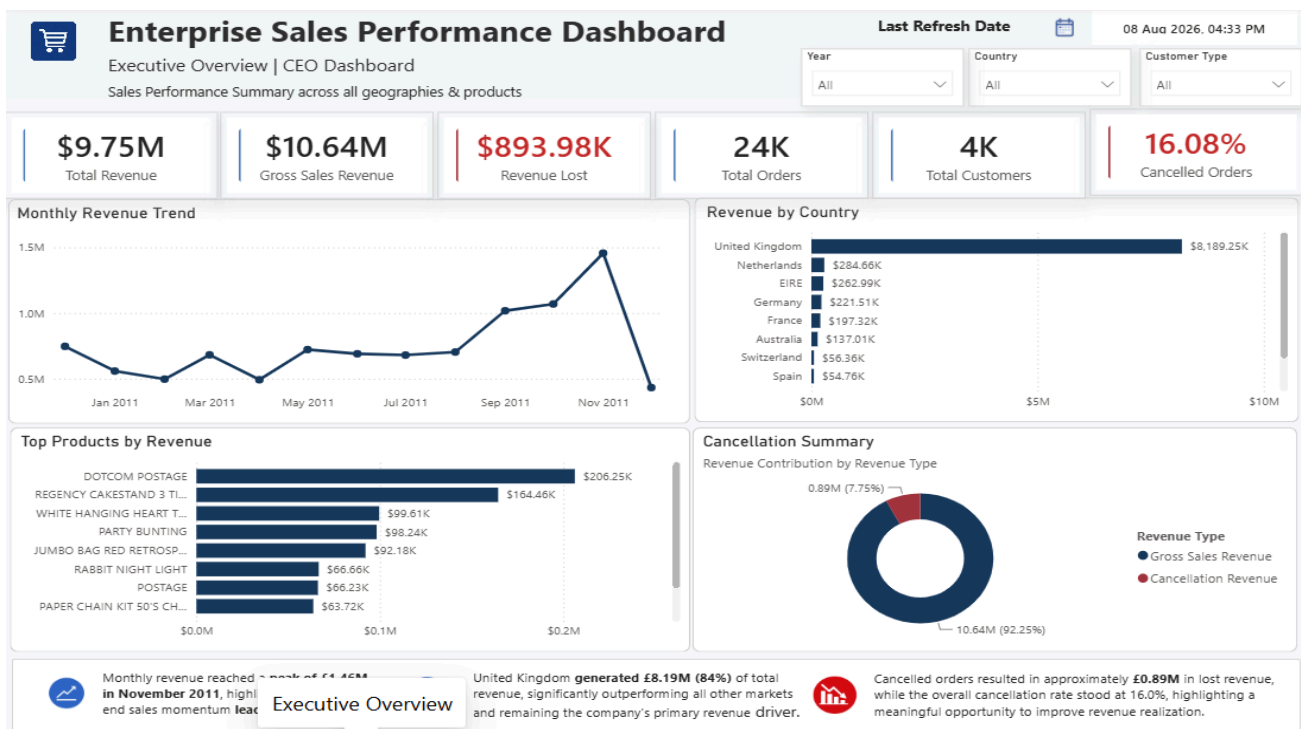
Customer Engagement

- Registered Customers make an average of 4.3 completed purchases, indicating that their lower order values are offset by stronger repeat purchasing behavior. Purchase frequency cannot be measured for Guest Customers because individual customers cannot be identified..

5. Executive Dashboards

The CEO dashboard focuses on overall business performance and provides insights into:

- Revenue performance
- Country performance
- Sales trends
- Product performance
- Cancellation and revenue-loss impact



CMO Dashboard

The CMO dashboard focuses on customer behavior and marketing performance, including:

- Customer segmentation
- Product popularity
- Customer mix and registration
- Repeat purchasing behavior
- Market engagement



Final Recommendations

- Diversify Market Exposure**
Expand selectively in high-performing international markets to reduce dependence on the UK.
- Prepare for Seasonal Demand**
Strengthen inventory, fulfilment, and staffing ahead of the year-end peak period.
- Strengthen Customer Retention**
Use personalized engagement and repeat-purchase initiatives for registered customers.
- Increase Customer Registration**
Convert high-value guest customers into registered customers to support long-term engagement.
- Optimize Product Planning**
Balance revenue contribution, sales volume, and customer demand when managing products.
- Reduce Revenue Leakage**
Investigate cancellation drivers and improve inventory and fulfilment processes.

7. Apply Customer-Specific Marketing

Use different strategies for guest conversion and registered customer retention.